



IN THE BICYCLE WORKSHOP

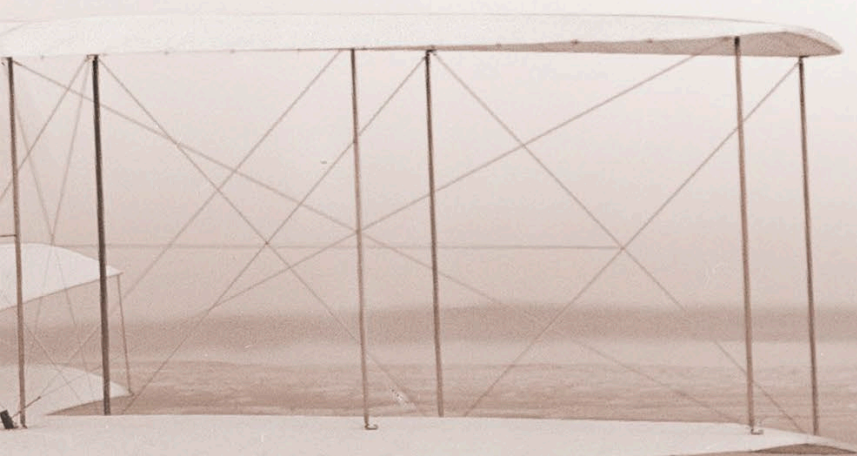
The Wright brothers began their working lives by repairing and making bicycles. They used these skills—and bicycle parts—to build their first plane, the *Flyer*. The aircraft had a wooden frame, and bicycle chains connected the plane's propellers to its engine.

THE GLIDER

Before they built their first powered plane, the Wright brothers experimented with wire-controlled gliders. They launched these gliders by running down sand dunes. The pilot steered the glider by moving a cradle attached to his hips, which then pulled wires to bend the wings to change the direction.



**Wright brothers
test a glider,
c.1901**



▲ THE FIRST FLIGHT

Orville Wright lies across the wing controls of the Flyer during its historic first flight, which lasted 12 seconds.

FLIGHT CONTROL

Orville Wright controlled *Flyer* by pulling wires to “warp,” or twist, the wings to lift one side or the other. This is called wing warping. Today, airplanes use hinged wing sections instead, but the idea of altering the wing shape to control a plane in the air was the big breakthrough.



1905

Orville makes a flight of 24 miles (38.9 km) in just 38 minutes at Huffman Prairie, Ohio, in a modified version of the *Flyer*.



1912

Wilbur dies at the age of 45 at the Wright family home on May 30, 1912. Orville becomes the president of the Wright company before selling it in 1915.

1948

Orville continues to work as a flight consultant for many decades. He dies on January 30, 1948, at the age of 70.